MAT354H1: Complex Analysis I

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The Algebra and Geometry of $\mathbb{C}$, Holomorphic Functions

Some information:

1. Instructor: Freid Tong
2. Lectures: Tuesday 1-3 PM, Thursday 12-1 PM
3. Tutorials: 2 sections, starts next week.
4. Textbook: Stein & Shakarchi, Complex Analysis
5. Grading: Homework: 10%, Midterm: 35%, Final 55 %

Complex analysis is the theory of calculus over the complex numbers, the complex numbers are given by the set

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R},\ i^2 = -1\}$$

At first it might seem like we are just replacing reals with complex, however, in this course we will see that there is more to this than replacing reals with complex numbers. The theory of complex analysis is richer than the reals. It is more geometric in nature. We will see that complex analysis has some rigid phenomena, there are some things in real analysis that do not happen in complex analysis, which we will see in this course.

We can perform the following operations on complex numbers:

1. Addition: If $z = a + bi$ and $w = c + di$ then $z + w = (a + c) + (b + d)i$
2. Multiplication: With the same set up, $z \cdot w = (a + bi)(c + di) = ac - bd + i(ad + bc)$

Remark: The set $\mathbb{C}$ with the addition and multiplication operations form a field.

A nice way to think of the complex numbers is by studying its geometry. The following figure shows us what happens in $\mathbb{C}$:

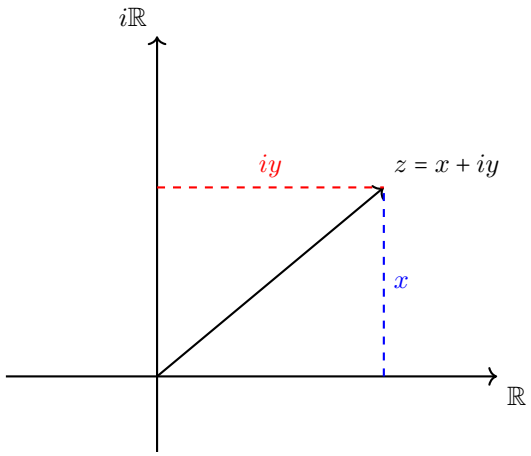


Figure 1: Geometric representation of $z = x + iy$.

So we can think of a complex number z as a vector where we have the real and imaginary parts of z, namely $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$. So when adding two complex numbers we are just performing vector addition, but we are not

multiplying in the same sense as the real numbers, but rather we can view $\mathbb{C}$ as a 2-dimensional vector space with additional structure.

Definition. Let $z \in \mathbb{C}$, we define the **complex conjugate** of z as the number $\bar{z} = x - iy$

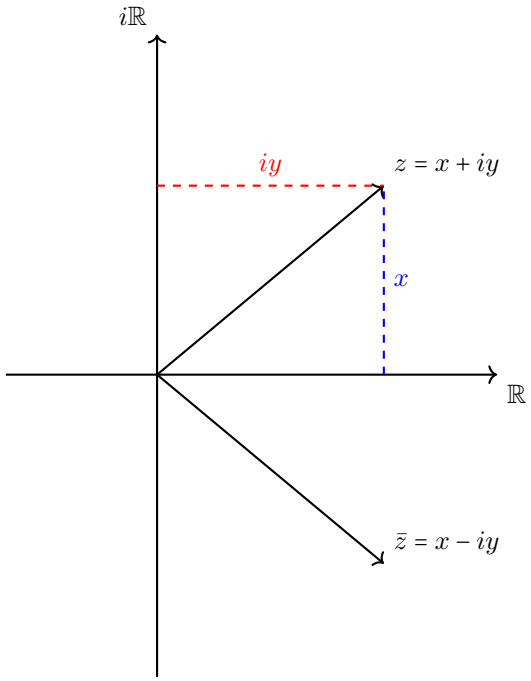


Figure 2: Representing the conjugate $\bar{z} = x - iy$.

So, going back to the graph, what we are doing is taking the number z and reflect it on the axis. From here we get the following quantity:

Definition. We define the **norm** of a complex number as

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}$$

In other words this is just the length of z

It turns out this conjugate is very useful for dividing complex numbers. Suppose that $z = x + iy$ is nonzero ($x, y \neq 0$). We want to write an expression for $\frac{1}{z}$. It turns out that this is given by

$$\frac{\bar{z}}{z\bar{z}} = \frac{x - iy}{x^2 + y^2} = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2}$$

From all of this, we have two viewpoints for complex numbers: Algebraic and Geometric This gives us a natural question: What is multiplication? Given $z \in \mathbb{C}$, we can view complex multiplication as a linear mapping $z\times : \mathbb{C} \to \mathbb{C}$. If $z \in \mathbb{R}$ then $z\times$ is just a dilation. The following figures show how this map acts with respect to the number z

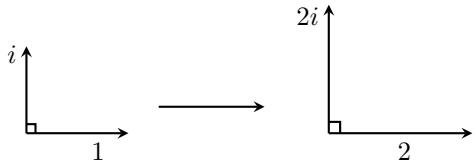


Figure 3: the mapping $2\times$

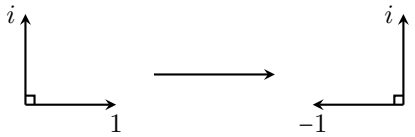


Figure 4: The mapping $i\times$

This tells us that complex multiplication is either a dilation or a rotation. Now to make our lives easier, let us introduce polar coordinates

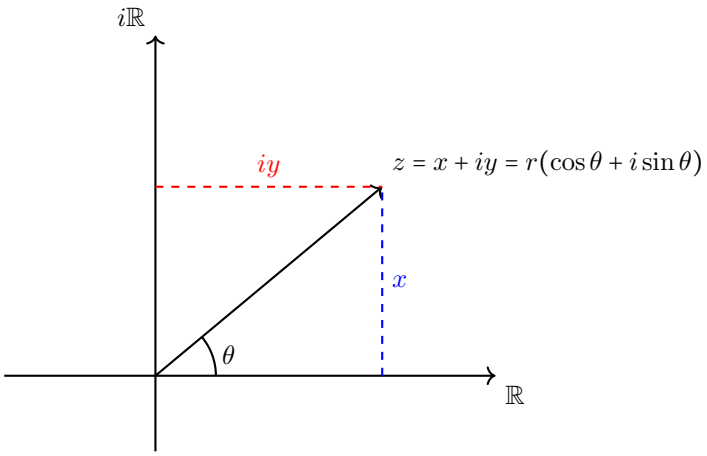


Figure 5: The polar form of z

In this case we have $z = r(\cos\theta + i\sin\theta)$ where r is the modulus of z and $(\cos\theta + i\sin\theta)$ a unit vector and θ is the **argument** of z or angle. Let us now do multiplication in polar coordinates

Example. Suppose $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ So now we have

$$\begin{aligned} z_1 \cdot z_2 &= r_1 r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \\ &= r_1 r_2(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i(\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2)) \\ &= r_1 r_2(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)) \end{aligned}$$

Remark: We are multiplying the norm, and adding the angles.

From now, we denote by $e^{i\theta}$ the expression

$$\cos\theta + i\sin\theta$$

Exercise: Verify $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$.

This is a very nice identity because it makes our life easier, and a lot of formulas sort of reduce into something nicer thanks to this, and later on we will see how this is useful in defining the complex exponential. So now we can write $re^{i\theta}$ instead of $r(\cos\theta + i\sin\theta)$.

Example. Let $T : \mathbb{C} \to \mathbb{C}$ be given by $z \mapsto (1+i)z$. This multiplies the length by $\sqrt{2}$ and rotate by $\frac{\pi}{4}$.

We can also think of these problems as a 2×2 matrix (since we can encode this as a linear transformation on $\mathbb{R}^2 \to \mathbb{R}^2$), so now we want to see which 2×2 matrix can represent this transformation. More generally, suppose that we have a linear map

$$T_w : \mathbb{C} \to \mathbb{C} \qquad z \mapsto w \cdot z$$

Again, we are multiplying the length by $|w|$ and rotate by $\arg(w)$. Now, if $w = a + bi$ and $z = x + yi$, then $T_w(x + yi) = ax - by + i(ay + bx)$. As a result, T_w is represented by the matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

One thing that we can conclude is that matrices of this type where we have equality along the diagonal and a sign change along the antidiagonal are ones that are produced by a rotation and a dilation

Now that we are done with the algebra and geometry of $\mathbb{C}$, we can move onto developing the calculus on $\mathbb{C}$. Suppose now that we have a function on some open set $\Omega \subseteq \mathbb{C} \to \mathbb{C}$, we want to talk about various notions, but first let us recall a few things from topology

$$\text{Topology of } \mathbb{C} = \text{Topology inherited from } \mathbb{R}^2$$

This topology is generated by open balls $D(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}$. This means that we have a notion of continuity, convergence, etc.

Definition. We say that $\lim_{z \to z_0} f(z) = L \in \mathbb{C}$ if

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t if } z \in D_\delta(z_0) \setminus \{z_0\} \implies |f(z) - L| < \varepsilon$$

Definition. Let $f : \Omega \to \mathbb{C}$, we say that f is **holomorphic** at a point $z_0 \in \Omega$ if the following limit exists:

$$\lim_{h \to 0,\ h \in \mathbb{C}} \frac{f(z_0 + h) - f(z_0)}{h}$$

If this limit exists, we denote it by $f'(z_0)$

Example. Consider $f(z_0) = z^n$, to see if this is holomorphic, we compute

$$\begin{aligned} f'(z_0) &= \lim_{h \to 0} \frac{(z+h)^n - z^n}{h} \\ &= \lim_{h \to 0} \frac{h}{h}\left((z+h)^{n-1} + (z+h)^{n-2} \cdot z + \cdots + (z+h) \cdot z^{n-2} + z^{n-1}\right) \\ &= nz^{n-1} \end{aligned}$$

Thus z^n is holomorphic on $\mathbb{C}$.

Example. Take $f(z) = \bar{z}$. As in the previous example, we have

$$\begin{aligned} f'(z) &= \lim_{h\to 0} \frac{(\overline{z+h}) - \overline{z}}{h} \\ &= \lim_{h\to 0} \frac{\bar{h}}{h} \end{aligned}$$

We run into a problem, as this limit does not exist due to the fact that the directional limits are not equal, one can take h real then h purely imaginary. This is an example of a function that is linear but not complex differentiable! If we were to view this as a mapping on $\mathbb{R}^2 \to \mathbb{R}^2$ via $(x, y) \mapsto (x, -y)$ which is real differentiable, but this is not holomorphic.

Theorem. If $f, g : \Omega \to \mathbb{C}$ are holomorphic at z_0, then

1. $(f+g)'(z_0) = f'(z_0) + g'(z_0)$
2. $(f \cdot g)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$
3. If $g(z_0) \neq 0$, then
$$\left(\frac{f}{g}\right)' = \frac{f'(z_0)g(z_0) - g'(z_0)f(z_0)}{g(z_0)^2}$$
Moreover, if $f : \Omega \to \mathbb{C}$ is holomorphic at z_0 and $g : U \to \mathbb{C}$ is holomorphic at $f(z_0)$ then $(g \circ f)'(z) = g'(f(z_0)) \cdot f'(z_0)$

Proof. Exercise. □

This implies that:

1. $f(z) = a_n z^n + \cdots + a_1 z + a_0$ is holomorphic.
2. Rational functions are holomorphic (be aware of the denominator), for instance, $f(z) = \frac{z+1}{z-i}$ is holomorphic on $\mathbb{C} \setminus \{i\}$.
3. $f(z) = |z|^2 = z \cdot \bar{z}$ is holomorphic at 0 (verify this!)

In light of these examples, a natural question to ask is: What is a holomorphic function? How does this complex derivative behave with respect to the real derivative? Now, we discuss the interaction of the $\mathbb{C}$-derivative and $\mathbb{R}$-derivative. To get started, let $f : \Omega \subseteq \mathbb{C} \to \mathbb{C}$ be written in terms of real valued functions u and v, that is $f = u + iv$. Now, let us write

$$F(x, y) = (u(x, y), v(x, y)) : \Omega \to \mathbb{R}^2$$

Recall (from 257/237) that F is differentiable at z_0 if and only if there exists a 2×2 matrix A such that

$$\lim_{|h|\to 0,\ h\in\mathbb{R}^2} \frac{|F(z_0 + h) - F(z_0) - A \cdot h|}{|h|} = 0$$

If the limit exists, then

$$A = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}(z_0)$$

One can immediately see that if $f'(z_0)$ exists, then

$$\lim_{h\to 0} \frac{f(z_0 + h) - f(z_0) - f'(z_0)h}{h} = 0$$

$$\implies \lim_{h \to 0} \frac{|f(z_0 + h) - f(z_0) - f'(z_0)h|}{|h|} = 0$$

So this implies that being complex differentiable/holomorphic $\implies$ real differentiable.

Cauchy-Riemann Equations and the Complex Exponential

Last time, we defined what it means for a function $f : \Omega \subseteq \mathbb{C} \to \mathbb{C}$ to be differentiable, namely the following limit exists:

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

Which looked very much the same as the definition for real derivatives, except h is now a complex number, so we take the limit across the complex plane. We also looked at some examples, such as polynomials, and some functions which were surprisingly not holomorphic (such $f(z) = \bar{z}$) anywhere, which was strange because if we view it as a real function it is differentiable but when switching to the complex perspective it is not. This naturally gives us the question "What does it mean for a real function $\mathbb{R}^2 \to \mathbb{R}^2$ to be holomorphic?"

Taking a coordinate $z = x + iy$, we write $f(z) = u(x, y) + iv(x, y)$, where u, v are both real valued functions on $\mathbb{R}^2$, and to avoid confusion we used the notation $F(x, y) = (u(x, y), v(x, y)) : \Omega \to \mathbb{R}^2$. F is real differentiable at z_0 if there is a 2×2 matrix A such that the following limit is 0.

$$\lim_{|h| \to 0} \frac{|F(z + h) - F(z) - A \cdot h|}{|h|}$$

Moreover, this matrix A is given by the partial derivatives, namely

$$A = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} (z_0) = DF(z_0)$$

It is easy to see that if f is $\mathbb{C}$-differentiable ($f'(z_0)$ exists) then $DF(z_0)$ exists. But then we notice that the real derivative (the matrix DF) is represented by 4 numbers, while the complex derivative is given (a priori) for one number which then can be written in 2 numbers.

So, one can take the limit

$$\begin{aligned} f'(z_0) &= \lim_{h \to 0,\ h \in \mathbb{R}} \frac{f(z_0 + h) - f(z_0)}{h} \\ &= \frac{\partial u}{\partial x}(z_0) + i \frac{\partial v}{\partial x}(z_0) \end{aligned}$$

So we are just taking the partial derivatives with respect to the real line, we can also write

$$\begin{aligned} f'(z_0) &= \lim_{h \to 0,\ h \in \mathbb{R}} \frac{f(z_0 + ih) - f(z_0)}{ih} \\ &= \frac{1}{i} \left(\frac{\partial u}{\partial y}(z_0) + i \frac{\partial v}{\partial y}(z_0) \right) \\ &= \frac{\partial v}{\partial y}(z_0) - i \frac{\partial u}{\partial y}(z_0) \end{aligned}$$

since $f'(z_0)$ exists (the limit defining it does not depend on the direction $h \to 0$ is taken from), both expressions for $f'(z_0)$ must agree:

$$\frac{\partial u}{\partial x}(z_0) + i \frac{\partial v}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) - i \frac{\partial u}{\partial y}(z_0)$$

Comparing real and imaginary parts of both sides gives

$$\frac{\partial u}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0), \qquad \frac{\partial v}{\partial x}(z_0) = -\frac{\partial u}{\partial y}(z_0)$$

So these constraints result into the function f satisfying the **Cauchy-Riemann** equations

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

Another interpretation would be that for a real differentiable function whose derivative is given by

$$DF = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}$$

Again, here we have a priori 2 inputs (u, v being the 2 real valued functions), then we have 4 numbers, so we can identify these equations via the matrix from last lecture (a rotation and a dilation)

$$DF = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

Theorem. $f : \Omega \to \mathbb{C}$ and $F : \Omega \to \mathbb{R}^2$ then $f'(z_0)$ exists if and only if $DF(z_0)$ exists and the Cauchy Riemann equations are satisfied

Proof. ($\Longrightarrow$) was already proven in the above discussion.

To prove ($\Longleftarrow$), note that $DF(z_0) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, then we have that

$$\lim_{h \to 0} \frac{\left| F(z+h) - F(z) - \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} h_x \\ h_y \end{pmatrix} \right|}{|h|} = 0$$

□

Example. We show that $f(z) = x^2 - y^2 + i2xy = z^2$ satisfies the Cauchy-Riemann equations. Indeed, note that $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$ then

$$\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y}$$

and

$$\frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}$$

Example. We show that $f(z) = \bar{z}$ does not satisfy the Cauchy-Riemann equations. Again, $\bar{z} = x - iy$ so $u(x, y) = x$ and $v(x, y) = -y$, then we get

$$\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = -1$$

and

$$\frac{\partial u}{\partial y} = 0 = \frac{\partial v}{\partial x}$$

hence $\bar{z}$ does not satisfy the Cauchy Riemann equations.

Let us now define the exponential. Fixing $z = x + iy$, we can extend the notion of the exponential in $\mathbb{R}$ to $\mathbb{C}$

Definition. The **complex exponential** is defined by $e^z = e^x(\cos y + i \sin y)$

We claim that this function satisfies the Cauchy-Riemann equations. To this end, we write $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$, then we have

$$\frac{\partial u}{\partial x} = e^x \cos y, \qquad \frac{\partial v}{\partial y} = e^x \cos y$$

and

$$\frac{\partial u}{\partial y} = -e^x \sin y, \qquad \frac{\partial v}{\partial x} = e^x \sin y$$

Therefore, the complex exponential satisfies the Cauchy-Riemann equations and it is also holomorphic, with $(e^z)' = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^z$.

Complex Line Integrals & Cauchy's Theorem

Last week, we derived the Cauchy-Riemann equations. More precisely, given $F : \Omega \to \mathbb{R}^2$, and $f : \Omega \to \mathbb{C}$. We proved that $f'(z_0)$ exists if and only if $DF(z_0)$ exists and the Cauchy-Riemann equations are satisfied, here DF is given by

$$DF(z_0) = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}(z_0)$$

In this lecture, we shift our focus towards integration of curves.

Definition. A **parametrized curve** in $\mathbb{C}$ is a continuous function $z(t) : [a, b] \to \mathbb{C}$.

1. $z(t)$ is **smooth** if $z'(t)$ exists and is nonzero and is also continuous.
2. $z(t)$ is **piecewise smooth** if there is a partition $a = a_1 < \cdots < a_n = b$ of $[a, b]$ such that $z(t)$ is smooth on $[a_k, a_{k+1}]$.

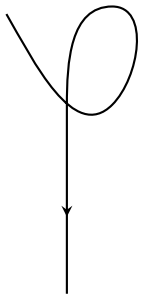


Figure 6: A curve

Two parameterizations $z(t) : [a, b] \to \mathbb{C}$ and $\tilde{z}(s) : [c, d] \to \mathbb{C}$ describe the same curve γ if $\tilde{z}(s) = z(t(s))$ for some $t(s) : [c, d] \to [a, b]$ which is C^1 and $t'(s) > 0$. This $t'(s)$ means a curve that comes with a "direction". See the following figure:

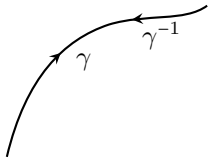
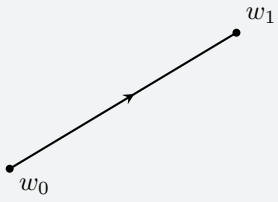


Figure 7: The curve γ and the reverse curve γ^-

Some times its useful to denote the reverse curve (basically the curve that is going in the opposite notation, we denote by γ^-). It is the same curve γ, but we have a reparameterization. In any case, this is the same definition used when studying line integrals.

Definition. A curve is **closed** if $z(a) = z(b)$. Moreover, a curve is **simple** if $z(c) \neq z(d)$ for $c \neq d \in [a, b]$ but $(c, d) \neq (a, b)$. If a curve starts and ends at the same point it is said to be a **simple closed curve**

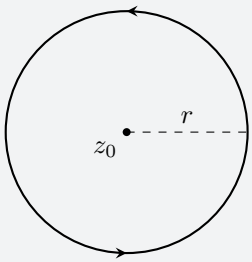
Example. Start at a point w_0 and stop at w_1, we can parametrize this using $z(t) = w_0 + t(w_1 - w_0)$ for $t \in [0, 1]$.



The derivative is given by $z'(t) = w_1 - w_0$.

Example. Consider a circle centered at z_0 with radius $r > 0$. We can parametrize this using

$$z(t) = z_0 + re^{it}, \quad t \in [0, 2\pi].$$



The derivative of this parametrization is given by $z'(t) = ire^{it}$.

Let $f : \Omega \to \mathbb{C}$ and $\gamma \in \Omega$ be a curve. We want to define the complex line integral

$$\int_\gamma f(z)\, dz$$

To make sense of this, we pick a parametrization $z(t) : [a, b] \to \Omega$, so the integral is given by

$$\int_a^b f(z(t))z'(t)dt$$

Suppose $f = u + iv$, we can write our integral as

$$\int_a^b \big(u(x(t), y(t)) + iv(x(t), y(t))\big)\big(x'(t) + iy'(t)\big)\, dt$$

It is not clear a priori why this is independent of the parametrization, because earlier we wrote down the line integral in terms of the curve and then we used the parametrization. The natural question rises: Why is this well-defined? Suppose $z(t) : [a, b] \to \Omega$ and $\tilde{z}(s) : [c, d] \to \Omega$. Then $\tilde{z}(s) = z(t(s))$. By the change of variables formula, we obtain the following

$$\int_c^d f(\tilde{z}(s))\tilde{z}'(s)ds = \int_c^d f(\tilde{z}(s))\tilde{z}'(s)\, ds = \int_c^d f\big(z(t(s))\big)\, z'(t(s))\, t'(s)\, ds = \int_a^b f(z(t))\, z'(t)\, dt,$$

Proposition. **(Properties of the Complex Line Integral)**

1. It is linear:
$$\int_\gamma (af(z) + bg(z))\, dz = a\int_\gamma f(z)\, dz + b\int_\gamma g(z)\, dz$$
2. $$\int_{\gamma^-} f(z)\, dz = -\int_\gamma f(z)\, dz$$
3. $$\left|\int_\gamma f(z\ )dz\right| \le \sup_{z\in\gamma} |f(z)| \text{Length}(\gamma).$$

Here the length of the curve γ is given by

$$\text{Length}(\gamma) = \int_a^b |z'(t)|\, dt$$

Proof. We prove (3). Indeed, suppose that we have a parametrization $z(t)$ of γ, then

$$\begin{aligned}\left|\int_a^b f(z(t))z'(t)\, dt\right| &\leq \int_a^b |f(z(t))||z'(t)|\, dt \\ &\leq \sup_{z\in\gamma}|f(z)|\int_a^b |z'(t)|dt \\ &= \sup_{z\in\gamma}|f(z)|\,\text{Length}(\gamma)\end{aligned}$$

□

Example. We compute $\int_\gamma z^n dz$ for $n \in \mathbb{Z}$, where γ denotes a circle centered at the origin with the positive orientation. To this end, we can parametrize this by $z(t) = re^{it}$ for $t \in [0, 2\pi]$. So

$$\begin{aligned}\int_\gamma z^n dz &= \int_0^{2\pi} i(re^{it})^{n+1}\, dt \\ &= ir^{n+1}\int_0^{2\pi} \cos((n+1)t) + i\sin((n+1)t)dt \\ &= \begin{cases} 0 & \text{if } n \neq -1 \\ 2\pi i & \text{if } n = -1 \end{cases}\end{aligned}$$

Example. We compute $\int_\gamma \bar{z}\, dz$ along the same curve as the previous example. If we use the same parametrization, then we have

$$\int_\gamma \bar{z}\, dz = \int_0^{2\pi} re^{-it}\cdot ire^{it}\, dt$$

Recall that

$$\begin{aligned}\overline{e^{it}} &= \overline{\cos t + i\sin t} \\ &= \cos t - i\sin t \\ &= \cos(-t) + i\sin(-t) \\ &= e^{-it}\end{aligned}$$

Now, going back to the integral, this simplifies to

$$\int_0^{2\pi} ir^2 dt = 2\pi i r^2$$

Now, going back to the first example, what makes the $n = -1$ special? is there a geometric reason? Suppose that $f : \Omega \to \mathbb{C}$ is a continuous, we say that $F : \Omega \to \mathbb{C}$ is a **primitive** of f if F is holomorphic and $F'(z) = f(z)$ for all $z \in \Omega$. One can think of this as the complex version of the real primitive.

Theorem. **(Fundamental Theorem of Contour Integrals)** Using the set up above. If $\gamma \subseteq \Omega$ is a curve, then

$$\int_\gamma f(z)dz = F(z(b)) - F(z(a))$$

Proof. Suppose that $z(t) : [a, b] \to \Omega$ is a parametrization of the curve γ. Then, if F is holomorphic we have

$$\begin{aligned} \frac{d}{dt}F(z(t)) &= F'(z(t))z'(t) \\ &= f(z(t))z'(t) && \text{(Since } F \text{ is a primitive of } f\text{)} \end{aligned}$$

By the Fundamental Theorem of Calculus, we have

$$\begin{aligned} \int_\gamma f(z)\ dz &= \int_a^b f(z(t))z'(t)dt \\ &= \int_a^b \frac{d}{dt}F(z(t))\ dt \\ &= F(z(b)) - F(z(a)). \end{aligned}$$

□

Example. Going back to the first integral we computed, we can use this fundamental theorem to compute

$$\int_\gamma z^n\ dz$$

Indeed, all we have to do is find a primitive, which in the $n \neq -1$ case, we have

$$\left(\frac{z^{n+1}}{n+1}\right)' = z^n$$

Corollary. If $\gamma \subseteq \Omega$ is a closed curve, and f has a primitive in Ω, then

$$\int_\gamma f(z)dz = 0$$

Corollary. If Ω is connected and $f : \Omega \to \mathbb{C}$ is holomorphic, and $f' \equiv 0$. Then f is constant.

It turns out if we take the integral of a holomorphic function along a closed curve, we often get zero. But in some cases, it is not, this will not be our focus now. The goal is to get to Cauchy's theorem.

Theorem. **(Cauchy's Theorem)** If $f : \mathbb{D} \to \mathbb{C}$ is a holomorphic function and γ is a simple closed curve whose interior is in $\mathbb{D}$. Then

$$\int_\gamma f(z)dz = 0$$

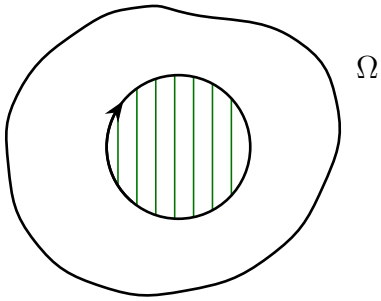


Figure 8: The region Ω with the curve γ in the interior

Due to time constraints we will not have the time to prove the full theorem, so we will prove a simpler version, due to Goursat

Theorem. **(Goursat's Theorem)** Let Ω be an open set in $\mathbb{C}$, $f : \Omega \to \mathbb{C}$ be a holomorphic function and let T be a triangle whose interior $\mathscr{T}$ is also in Ω. Then

$$\int_T f(z)dz = 0$$

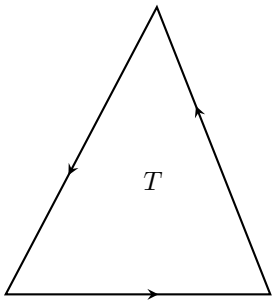


Figure 9: The boundary of the triangle T

Proof. Next time! □

Proof of Goursat's Theorem

Last time, we defined what a complex line integral is. Given a continuous $f : \Omega \to \mathbb{C}$ for Ω an open set in $\mathbb{C}$, the integral of f along a curve $\gamma \subseteq \Omega$ is given by

$$\int_\gamma f(z)\ dz := \int_a^b f(z(t))z'(t)\ dt$$

We also showed that this is independent of the parametrization. There is another way to see it, essentially we can write it as the sum of 2 real line integrals (which again, are independent), so dz here is some complex line element, given by $dz = dx + idy$, so when we take the complex valued function, multiplying it by the line element dz. In other words,

$$\int_\gamma (u + iv)(dx + idy)$$

We then computed a bunch of line integrals and showed that the complex line integral satisfies some basic properties (linearity, etc) and proved the Fundamental Theorem of Calculus for Complex Line integrals (otherwise known as the Fundamental theorem of Contour Integrals).

Theorem. If f has a primitive $F(z) : \Omega \to \mathbb{C}$. That is, to say $F'(z) = f(z)$. Then

$$\int_\gamma f(z)dz = F(z(b)) - F(z(a))$$

A corollary of this theorem is that if f has a primitive and we integrate along a closed curve, that integral must be zero, i.e.

$$\int_\gamma f(z) = 0$$

for γ a closed curve. Which makes sense because the end points are equivalent, so by the fundamental theorem the result follows. We can do more, last time, we stated a theorem due to Goursat

Theorem. **(Goursat's Theorem)** Suppose that $\overline{\mathscr{T}} \subseteq \Omega$ is a triangle and $T = \partial\mathscr{T}$ is the boundary. If $f : \Omega \supseteq \overline{\mathscr{T}} \to \mathbb{C}$ is holomorphic, then

$$\int_T f(z)\ dz = 0$$

Proof. The idea is to use subdivision of the triangle into smaller ones, indeed. Let $T^{(0)} = T$, and let $L = \text{length}(T^{(0)})$ and $d = \text{diam}(\overline{\mathscr{T}}^{(0)})$. Bisect each side of $\mathscr{T}^{(0)}$ at its midpoint and connect the three midpoints. This partitions $\mathscr{T}^{(0)}$ into four smaller triangles

$$\mathscr{T}_1^{(0)}, \mathscr{T}_2^{(0)}, \mathscr{T}_3^{(0)}, \mathscr{T}_4^{(0)},$$

each similar to $\mathscr{T}^{(0)}$ with sides half as long. Orient each boundary $\partial\mathscr{T}_k^{(0)}$ positively (counterclockwise).

Now summing the four integrals gives back the original

$$\int_{T^{(0)}} f(z)\,dz = \sum_{k=1}^{4} \int_{\partial\mathscr{T}_k^{(0)}} f(z)\,dz.$$

Indeed, every edge of the subdivision is either (i) part of the boundary of $\mathscr{T}^{(0)}$, traversed once with the correct orientation, or (ii) an interior segment (connecting two midpoints), shared by exactly two of the small triangles and traversed once in each direction. The interior contributions cancel pairwise, leaving exactly the integral over $T^{(0)}$.

By the triangle inequality,

$$\left|\int_{T^{(0)}} f(z)\,dz\right| \le \sum_{k=1}^{4} \left|\int_{\partial \mathscr{T}_k^{(0)}} f(z)\,dz\right|,$$

so at least one $k \in \{1,2,3,4\}$ satisfies

$$\left|\int_{\partial \mathscr{T}_k^{(0)}} f(z)\,dz\right| \ge \frac{1}{4}\left|\int_{T^{(0)}} f(z)\,dz\right|.$$

Let us call this triangle $T^{(1)}$. We repeat the subdivision on $T^{(1)}$, and choose a sub-triangle $T^{(2)}$ with the analogous property, and so on. This produces a nested sequence of closed triangles

$$\overline{\mathscr{T}}^{(0)} \supseteq \overline{\mathscr{T}}^{(1)} \supseteq \overline{\mathscr{T}}^{(2)} \supseteq \dots$$

such that for every $n \ge 0$:

$$\left|\int_{T^{(n)}} f(z)\,dz\right| \ge \frac{1}{4^n}\left|\int_{T^{(0)}} f(z)\,dz\right|, \tag{$*$}$$

$$\operatorname{diam}\left(\overline{\mathscr{T}}^{(n)}\right) = \frac{d}{2^n}, \qquad \operatorname{length}\left(T^{(n)}\right) = \frac{L}{2^n}.$$

Note that the sets $\overline{\mathscr{T}}^{(n)}$ are nonempty, compact, nested, and $\operatorname{diam}(\overline{\mathscr{T}}^{(n)}) \to 0$. By the nested compact set theorem, there exists a unique point

$$z_0 \in \bigcap_{n=0}^{\infty} \overline{\mathscr{T}}^{(n)} \subseteq \overline{\mathscr{T}}^{(0)} \subseteq \Omega.$$

Claim:

$$\left|\int_{T^{(k)}} f(z)\ dz\right| \le \frac{1}{4^k}\varepsilon^k$$

Such that $\varepsilon_k \to 0$ whenever $k \to \infty$.

Proof. Since f is holomorphic at z_0, let us define

$$\psi(z) = \begin{cases} \dfrac{f(z) - f(z_0) - f'(z_0)(z - z_0)}{z - z_0}, & z \ne z_0, \\ 0, & z = z_0. \end{cases}$$

By definition of $f'(z_0)$, ψ is continuous at z_0 with $\psi(z_0) = 0$; that is,

$$\psi(z) \to 0 \quad \text{as } z \to z_0.$$

For $z \ne z_0$ this rearranges to

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0),$$

and this identity also holds trivially at $z = z_0$. We show that the linear part vanishes. Indeed, the function $z \mapsto f(z_0) + f'(z_0)(z - z_0)$ has the primitive $F(z) = f(z_0)z + \frac{1}{2}f'(z_0)(z - z_0)^2$ on all of $\mathbb{C}$. Since $T^{(k)}$ is a closed curve,

$$\int_{T^{(k)}} \left[f(z_0) + f'(z_0)(z - z_0)\right] dz = 0,$$

so

$$\int_{T^{(k)}} f(z)\,dz = \int_{T^{(k)}} \psi(z)(z - z_0)\,dz.$$

Now we set

$$\varepsilon_k := d\,L \cdot \sup_{z \in \overline{\mathscr{T}}^{(k)}} |\psi(z)|.$$

This is well-defined since ψ is continuous on the compact set $\overline{\mathscr{T}}^{(k)}$ (continuous away from z_0, and continuous at z_0 by the differentiability of f at that point). We show that $\varepsilon_k \to 0$. Given $\eta > 0$, continuity of ψ at z_0 provides $\delta > 0$ such that $|\psi(z)| < \eta$ whenever $|z - z_0| < \delta$. Since $z_0 \in \overline{\mathscr{T}}^{(k)}$ for every k and $\operatorname{diam}(\overline{\mathscr{T}}^{(k)}) = d/2^k \to 0$, there is N such that for all $k \geq N$,

$$\overline{\mathscr{T}}^{(k)} \subseteq D(z_0, \delta),$$

which forces $\sup_{z \in \overline{\mathscr{T}}^{(k)}} |\psi(z)| \leq \eta$, hence $\varepsilon_k \leq dL\,\eta$ for $k \geq N$. As $\eta > 0$ was arbitrary, $\varepsilon_k \to 0$.

Now, for $z \in T^{(k)} \subseteq \overline{\mathscr{T}}^{(k)}$, we have $|z - z_0| \leq \operatorname{diam}(\overline{\mathscr{T}}^{(k)}) = d/2^k$ and $|\psi(z)| \leq \sup_{\overline{\mathscr{T}}^{(k)}} |\psi|$. By the properties of the complex line integral, we have

$$\left| \int_{T^{(k)}} f(z)\,dz \right| = \left| \int_{T^{(k)}} \psi(z)(z - z_0)\,dz \right| \leq \Big(\sup_{\overline{\mathscr{T}}^{(k)}} |\psi| \Big) \cdot \frac{d}{2^k} \cdot \operatorname{length}(T^{(k)}) = \Big(\sup_{\overline{\mathscr{T}}^{(k)}} |\psi| \Big) \cdot \frac{d}{2^k} \cdot \frac{L}{2^k}.$$

That is,

$$\left| \int_{T^{(k)}} f(z)\,dz \right| \leq \frac{dL \cdot \sup_{\overline{\mathscr{T}}^{(k)}} |\psi|}{4^k} = \frac{\varepsilon_k}{4^k},$$

which is exactly desired inequality, with $\varepsilon_k \to 0$ as $k \to \infty$. □

Combining the claim with $(*)$, we obtain, for every k,

$$\left| \int_{T^{(0)}} f(z)\,dz \right| \leq 4^k \left| \int_{T^{(k)}} f(z)\,dz \right| \leq 4^k \cdot \frac{\varepsilon_k}{4^k} = \varepsilon_k.$$

Letting $k \to \infty$, since $\varepsilon_k \to 0$, we conclude

$$\left| \int_T f(z)\,dz \right| = 0,$$

namely

$$\int_T f(z)\,dz = 0,$$

as required. □

Cauchy's Theorem

Last time, we defined what a complex line integrals, and proved the Fundamental Theorem of Calculus for Line Integrals (also known as the Fundamental theorem of line integrals). If $f : \Omega \to \mathbb{C}$ has a primitive $F(z) : \Omega \to \mathbb{C}$. That is, to say $F'(z) = f(z)$. Then

$$\int_\gamma f(z)dz = F(z(b)) - F(z(a))$$

Which is a consequence of the Fundamental theorem of calculus for real integrals. Moreover, if $f : \Omega \to \mathbb{C}$ has a primitive then

$$\int_\gamma f(z) = 0$$

For any closed $\gamma \subseteq \Omega$. Then we proved Goursat's theorem, which states that for $f : \Omega \to \mathbb{C}$ a holomorphic function and T is a triangle with its interior in Ω, then

$$\int_T f(z)\ dz = 0$$

We state a theorem of which Goursat's theorem is a special case.

Theorem. (Cauchy's theorem in a disk) Suppose that $f : \mathbb{D} \to \mathbb{C}$ is a holomorphic function. Then f has a primitive in $\mathbb{D}$. In particular, this implies that

$$\int_\gamma f(z)\ dz = 0$$

For every closed curve $\gamma \subseteq \mathbb{D}$.

Example. (A holomorphic function without a primitive) Take $\frac{1}{z} : \mathbb{C}^* \to \mathbb{C}$, we previously showed that

$$\int_{C_r} \frac{dz}{z} = 2\pi i \neq 0$$

Proof of Cauchy's Theorem. The idea is to construct a primitive of f. Given a point z, we construct a path $0 \mapsto z$, let us call it γ_z. Given $z + h$, we construct γ_{z+h} by first following γ_z from $0 \mapsto z$, and then following the straight line segment from $z \mapsto z + h$ where $h \in \mathbb{C}$ is small enough, denoted by $\gamma_{[z,z+h]}$. By additivity of line integrals, one has that

$$\int_{\gamma_{z+h}} f(w)\, dw = \int_{\gamma_z} f(w)\, dw + \int_{\gamma_{[z,z+h]}} f(w)\, dw.$$

If we were to have a primitive it would be given by

$$F(z) = \int_{\gamma_z} f(w)\, dw$$

Claim: $F'(z) = f(z)$

Proof of claim: We compute the complex derivative of F. To this end,

$$\begin{aligned} \frac{F(z+h) - F(z)}{h} &= \frac{\int_{\gamma_{z+h}} f(w)\ dw - \int_{\gamma_z} f(w)\ dw}{h} \\ &= \frac{\int_{\gamma_{[z,z+h]}} f(w)\ dw}{h} \end{aligned} \qquad \text{(Goursat's theorem)}$$

This implies that

$$\left|\frac{F(z+h)-F(z)}{h}-f(z)\right| = \left|\frac{\int_{\gamma_{z+h}} f(w)\ dw - \int_{\gamma_z} f(w)\ dw}{h}\right|$$
$$\leq \sup_{w\in\gamma_{[z,z+h]}} |f(w)-f(z)| \to 0$$

as $h \to 0$. □

Which proves the theorem.

□

Now, one might be wondering why does this not work for $\frac{1}{z}$. The thing is, it is not defined on 0, but later we will see how can it have a primitive. More on this when we study poles. In the proof of Cauchy's theorem, we used a particular property of disks that allowed us to construct the path, that is convexity, we realize that path connectedness is not enough due to the fact that $\frac{1}{z}$ does not have a primitive on $\mathbb{C}^*$ and it is a path connected set.

Remark: Goursat's theorem still holds if we replace triangles by polygonal domains. By polygonal domain, we mean a domain that is a polygon (not necessarily convex). The observation we can make is that if we have a polygonal domain, we can decompose into a union of triangles. So, since Goursat's theorem holds for triangle, it must hold for such domains. See this week's homework for more details.

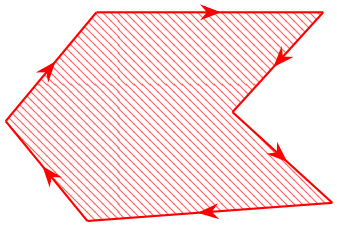


Figure 10: A polygon

Note that the same thing holds if $\mathbb{D}$ is replaced by a convex open set. The proof is the same. Cauchy's theorem holds in more generality. This generality will be on a **toy contour**, namely a simple closed curve with an interior.

Example. Any convex set.

Example.

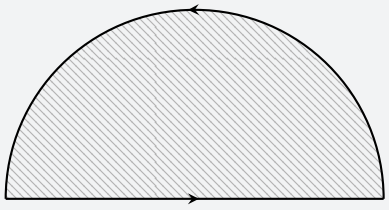


Figure 11: A semicircle

Example.

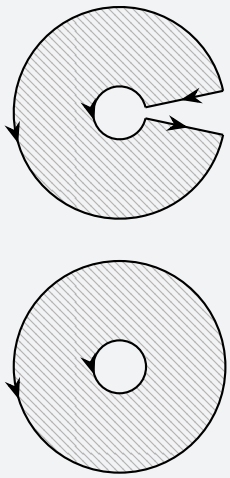


Figure 12: Some examples of keyhole contours, which are going to be very important for this course.

We don't really care about the definition of the interior here, but for anyone who is interested, see the Jordan curve theorem. This theorem states that every closed curve has a well defined interior that is simply connected. Now the claim is that Cauchy's theorem extends to these regions.

Theorem. (**Cauchy's theorem**) Suppose that $f : \Omega \to \mathbb{C}$ is a holomorphic function. Then f has a primitive in Ω. In particular, this implies that

$$\int_\gamma f(z)\ dz = 0$$

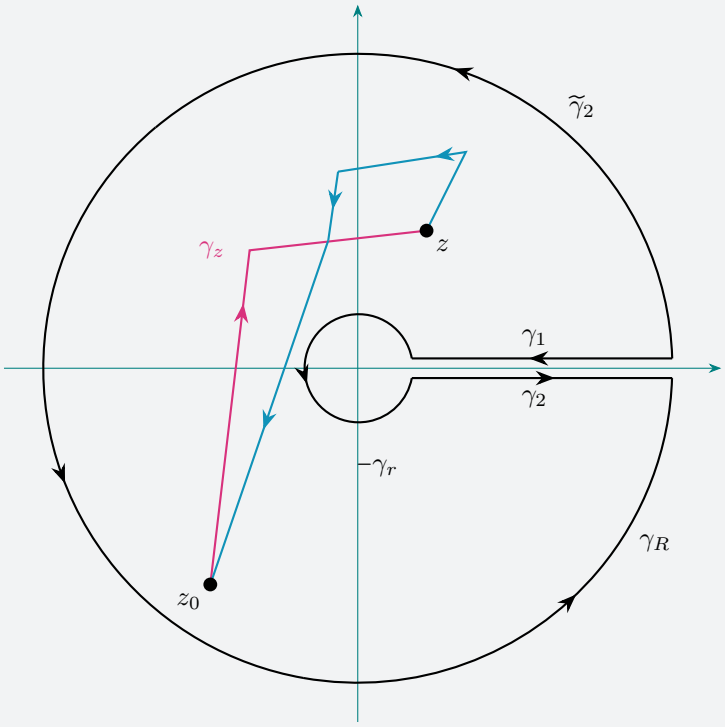
For every closed curve $\gamma \subseteq \Omega$.

The proof idea is roughly the same, but now we can use Goursat's theorem on polygonal domains.

Example. Suppose that $f(z) = \frac{1}{z}$, this is defined everywhere except at the origin. We can draw a toy contour where the outer one goes along the circle of radius R, we denote this by C_R, and the inner one goes along a circle of radius r, we also denote this by C_r. We imitate the same process as before, fix a point z_0 and another point z. Define

$$F(z) = \int_{\gamma_{z_0}} f(w)\, dw$$

We choose a piecewise curve that connects these two points and we will choose another one. We claim that it does not matter which curve do we end up choosing. To this end, note that we can write this path as a sum of polygons. The point is, if we pick the domain to be something like an annuli, then we know that $1/z$ does not have a primitive, and so this construction goes wrong.



Using some topology, we can say that one of the sets is simply-connected and the other is not. Furthermore, a similar argument shows that $F'(z) = f(z)$. Now, by Cauchy's theorem, it follows that

$$\int_{\gamma_R} \frac{dz}{z} - \int_{\gamma_r} \frac{dz}{z} + \int_{\gamma_1} \frac{dz}{z} + \int_{\gamma_2} \frac{dz}{z} = 0$$

Here γ_1 and γ_2 are the two segments connecting the inner circle C_r to the outer circle C_R, separated by a thin corridor of width $\varepsilon > 0$ that we introduce so the contour avoids self-intersection. As $\varepsilon \to 0$, the two segments γ_1 and γ_2 come together and their contributions to the integral cancel, which leaves only the integrals over C_r and C_R. Now, letting $\varepsilon \to 0$, we get

$$\int_{C_r} \frac{dz}{z} = \int_{C_R} \frac{dz}{z} = 0$$

implying that

$$\int_{C_r} \frac{dz}{z} = \int_{C_R} \frac{dz}{z}$$

Turns out that these ideas are very powerful. The above example is the most basic use of contour integration. We will see more examples of how to use these contours to evaluate integrals via Cauchy's theorem. We can use this to compute some hard integrals, but our life would be easy by using contour integration. Some of which arise from the theory of Fourier transform

Example. We compute

$$\int_{-\infty}^{\infty} e^{-\pi x^2 - 2\pi i x\xi}\, dx = h(\xi)$$

It is known that $h(1)$ is the Gaussian integral. To this end, it turns out that the right way to think of this, is we extend the function to a function on the complex plane, so we write $f(z) = e^{-\pi z^2}$, this is holomorphic in $\mathbb{C}$.

The claim is that this integral can be expressed as a line integral. Namely

$$\int_{\gamma_z} f(z)\ dz$$

Indeed, take $z(t) = t + i\xi$, our integral can now be written as

$$\begin{aligned}\int_{\gamma_z} f(z)\ dz &= \int_{-\infty}^{\infty} e^{-\pi(t+i\xi)^2}\ dt \\ &= \int_{-\infty}^{\infty} e^{-\pi t^2 - 2\pi i\xi t + \pi\xi^2}\ dt \\ &= h(\xi)e^{\pi\xi^2}\end{aligned}$$

Now, we can apply Cauchy's theorem, where we can integrate along the curve

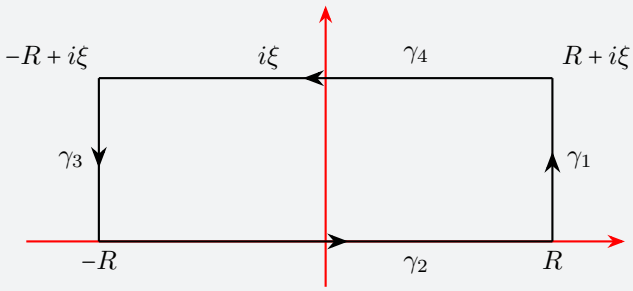


Figure 13: The contour we are integrating over.

Indeed, we have

$$\int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz + \int_{\gamma_3} f(z)\,dz + \int_{\gamma_4} f(z)\,dz = 0,$$

implying that

$$\int_{\gamma_4} f(z)\,dz = -\int_{\gamma_1} f(z)\,dz - \int_{\gamma_2} f(z)\,dz - \int_{\gamma_3} f(z)\,dz.$$

Note that

$$\int_{\gamma_2} f(z)\,dz = \int_{-R}^{R} e^{-\pi t^2}\,dt$$

and

$$\int_{\gamma_3} f(z)\,dz = \int_{0}^{\xi} e^{-\pi(-R+it)^2}\,i\,dt.$$

For γ_1, we can take the parametrization $z(t) = R + it$, to get

$$\int_{\gamma_1} f(z)\,dz = i\int_{0}^{\xi} e^{-\pi R^2}\,e^{-2\pi i R t}\,e^{\pi t^2}\,dt.$$

Moreover, we have

$$\left|\int_{\gamma_1} f(z)\,dz\right| \le e^{-\pi R^2}\,|\xi|\,e^{\pi\xi^2}.$$

A similar argument holds for γ_3. Now, we get

$$\begin{aligned} h(\xi)e^{\pi\xi^2} &= \lim_{R\to\infty}\int_{\gamma_4} f(z)\,dz \\ &= -\lim_{R\to\infty}\Big(\int_{\gamma_1} f(z)\,dz + \int_{\gamma_3} f(z)\,dz\Big) - \lim_{R\to\infty}\int_{\gamma_2} f(z)\,dz \\ &= 1. \end{aligned}$$

Another Integral & Cauchy's Integral Formula

Last time, we talked about Cauchy's theorem, which says for the interior of a simple closed curve Ω, and $f : \Omega \to \mathbb{C}$ a holomorphic function. Then f has a primitive in Ω and

$$\int_\gamma f(z)\ dz = 0$$

For any closed curve $\gamma \subseteq \Omega$. And then we saw one example where we showed that the Fourier transform of a Gaussian is itself. So contour integration is very useful for evaluating some harder definite integrals, where we can just elevate to the complex plane and cleverly apply Cauchy's theorem.

"These things are like art. There is no Algorithm" - Freid Tong

Example. We use contour integration to evaluate

$$\int_0^\infty \frac{1-\cos x}{x^2}\ dx \tag{✯}$$

So, we want to elevate this to a function on $\mathbb{C}$ and then write it as a line integral. Recall that $e^z = e^x(\cos y + i \sin y)$. This suggests that if we were to evaluate a line integral, where this appears. Indeed, consider the function

$$f(z) = \frac{1-e^{iz}}{z^2}.$$

This is holomorphic in $\mathbb{C} \setminus 0$ since it is the quotient of 2 holomorphic functions and it is well defined on $\mathbb{C} \setminus 0$. The claim is that the real part of the integral along the real axis will give us (✯) in the limit. The contour we will be using is an indented semicircle with the smaller semicircle having radius ε near the origin with clockwise orientation and the larger semicircle with radius R with the counterclockwise orientation.

So Cauchy's theorem tells us that the integral along the boundary is 0. So, we have to integrate along γ_ε and γ_R. More precisely,

$$\int_{\gamma_1} + \int_{\gamma_2} = -\int_{\gamma_\varepsilon} - \int_{\gamma_R}.$$

We start by computing

$$\int_{\gamma_R} \frac{1-e^{iz}}{z^2}\ dz.$$

First, note that the norm of the numerator is bounded and the norm of the denominator grows like R^2, so we use the length estimate to obtain

$$\left| \int_{\gamma_R} \frac{1-e^{iz}}{z^2}\ dz \right| \le \text{Length}(\gamma_R) \sup_{\gamma_R} \frac{|1-e^{iz}|}{|z^2|} \le \pi R \frac{2}{R^2} = \frac{2\pi}{R} \to 0$$

as $R \to \infty$. Now, we evaluate the integral along γ_ε, this is a little more involved. We claim that for $|z| \ll 1$ we have that

$$e^z = 1 + z + O(|z|^2)$$

(we may believe this for now, we will come back to this when we discuss power series). So this tells us that

$$e^{iz} = 1 + iz + O(|z|^2).$$

When we plug that into the integral, we get

$$\int_{\gamma_\varepsilon} \frac{1-e^{iz}}{z^2}\ dz = \int_{\gamma_\varepsilon} -\frac{iz}{z^2}\ dz + \int_{\gamma_\varepsilon} \frac{O(|z|^2)}{z^2}\ dz.$$

Note that the second term is $o(\varepsilon)$. Now, we take a parameterization $z(t) = \varepsilon e^{-it}$, so that

$$z'(t) = -i\varepsilon e^{-it}$$

Thus

$$\begin{aligned}\int_{\gamma_\varepsilon} \frac{1-e^{iz}}{z^2}\,dz &= -i\int_{\gamma_\varepsilon}\frac{dz}{z} + o(\varepsilon)\\ &= -i\int_{\pi}^{2\pi}\frac{1}{\varepsilon e^{-it}}(-i)\varepsilon e^{-it}\,dt + o(\varepsilon)\\ &= -\pi + o(\varepsilon).\end{aligned}$$

Therefore,

$$\lim_{\varepsilon\to 0}\int_{\gamma_\varepsilon}\frac{1-e^{iz}}{z^2}\,dz = -\pi.$$

This implies that

$$\lim_{\substack{\varepsilon\to 0\\ R\to\infty}}\left(\int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz\right) = \pi.$$

Taking the real part, we have

$$\begin{aligned}\pi &= \lim_{\substack{\varepsilon\to 0\\ R\to\infty}} \operatorname{Re}\left(\int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz\right)\\ &= \lim_{\substack{\varepsilon\to 0\\ R\to\infty}}\left(\int_{\varepsilon}^{R}\frac{1-\cos x}{x^2}\,dx + \int_{-R}^{-\varepsilon}\frac{1-\cos x}{x^2}\,dx\right)\\ &= 2\int_0^\infty \frac{1-\cos x}{x^2}\,dx.\end{aligned}$$

Hence

$$\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}.$$

Now, we will start talking about another consequence of Cauchy's theorem, namely Cauchy's Integral Formula.

Theorem. (Cauchy's Integral Formula) Suppose that $f:\Omega\to\mathbb{C}$ is holomorphic, and $\overline{D_r(z_0)}\subseteq\Omega$. Then

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta$$

Where $C = \partial D_r(z_0)$ with the counterclockwise orientation and $z \in D_r(z_0)$.

The idea behind the proof is quite simple. From Cauchy's theorem, we want to evaluate the integral

$$\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \tag{\#}$$

This should not depend on the circle C as long as z is in the interior. Suppose that there is a single pole at z. the claim is that if we integrate over this circle and then over another circle, the integrals must be equal! Since we can connect a small circle around z to C with a corridor of width $\delta > 0$, we obtain a toy contour on which $f(\zeta)/(\zeta - z)$ is holomorphic (since we excluded the singularity at z by the smaller circle). By Cauchy's theorem, the integral over this toy contour is 0. As the corridor width $\delta \to 0$, the two connecting segments cancel, and so the integrals over the larger and smaller circles agree. To be continued next week.

Proof of Cauchy's Integral Formula and Some Consequences

Last time, we started discussing Cauchy's integral formula. So this is where we will start today. This is one of the most important results in complex analysis, and it will tell us everything that we need to know about holomorphic functions. Recall that for f a holomorphic function on $\Omega \to \mathbb{C}$. $\overline{D_r(z_0)} \subseteq \Omega$. Then for any $z \in D_r(z_0)$

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Where C is the boundary of $D_r(z_0)$ with the counterclockwise orientation.

There are many implications of this result, and we will see quite a few, if you notice, on the left side we have a holomorphic function, but on the right side, we have an integral, and we have a term $\zeta - z$, which is smooth, so it turns out that being holomorphic is more than just having one derivative. In fact, holomorphic functions are infinitely differentiable.

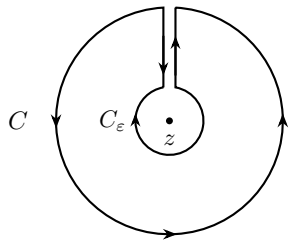


Figure 14: The keyhole contour inside we are integrating over, This consists of an outer circle C, and a smaller circle C_ε with radius ε, the corridor here has width δ.

Proof of Cauchy's Integral Formula. By Cauchy's theorem, we want to show that

$$\frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \int_{C_\varepsilon(z)} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Now, we want to evaluate the integral on the RHS. To this end, we parametrize $C_\varepsilon(z)$ by $\zeta(t) = z + \varepsilon e^{it}$. This implies that the right hand side

$$\frac{1}{2\pi i} \int_0^{2\pi} \varepsilon \frac{f(\zeta(t))}{\varepsilon e^{it}} e^{it} \, dt = \frac{1}{2\pi i} \int_0^{2\pi} f(z + \varepsilon e^{it}) \, dt \tag{$\bigstar$}$$

Note that ($\bigstar$) is the mean value formula. So we are averaging around the parametrized curve. So now, if $\varepsilon \to 0$, this becomes

$$\frac{1}{2\pi} \int_0^{2\pi} f(z) \, dz = f(z).$$

□

Example. We compute the integral

$$\int_C \frac{1}{z} \, dz$$

Where C is the circle containing 0 in the interior. By Cauchy's integral formula, it follows that this is just $2\pi i$.

Example. Suppose that C = circle of radius 2 about the origin. We compute

$$\int_C \frac{\zeta^2+1}{\zeta-1}\,d\zeta$$

Cauchy's integral formula (applied to f at $f(\zeta)=\zeta^2+1$, evaluated at 1) tells us that

$$\int_C \frac{\zeta^2+1}{\zeta-1}\,d\zeta = 2\pi i f(1) = 4\pi i$$

Example. We consider the same curve as the previous example, but this time we are integrating

$$\int_C \frac{\zeta^2+1}{\zeta-3}\,d\zeta$$

We get 0 by Cauchy's theorem, since 3 is not inside the disc we are integrating over.

We state another consequence of Cauchy's integral formula.

Proposition. (Mean Value Identity) If f is holomorphic on Ω an open set containing $\overline{D_r(z_0)}$, then

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0+re^{it})\,dt$$

This is an immediate consequence of CIF, by taking $C=\partial D_r(z_0)$. Basically it says that f is always equal to the average value around a circle. Note that this does not hold for $\mathbb{R}$-differentiable functions, even smooth functions. Another consequence that was hinted at earlier in the lecture, was that f can be represented by an integral, where the dependence is on ζ, moreover, it turns out that we can differentiate infinitely many times.

To prove this, differentiate the Cauchy integral formula to obtain

$$\begin{aligned} f'(z) &= \lim_{h\to 0}\frac{f(z+h)-f(z)}{h} \\ &= \lim_{h\to 0}\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{h}\left(\frac{1}{\zeta-z-h}-\frac{1}{\zeta-z}\right)d\zeta \\ &= \lim_{h\to 0}\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta-z-h)(\zeta-z)}\,d\zeta. \end{aligned}$$

Since z lies in the interior of C, we have

$$d = \min_{\zeta\in C}|\zeta-z| > 0.$$

For $|h| < d/2$, the denominator $\zeta-z-h$ is bounded away from zero on C. Moreover,

$$\frac{1}{(\zeta-z-h)(\zeta-z)} \rightrightarrows \frac{1}{(\zeta-z)^2}$$

for $\zeta\in C$ as $h\to 0$. so by uniform convergence, we can interchange the limit and the integral. Hence,

$$f'(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta-z)^2}\,d\zeta.$$

We now differentiate again to get

$$\begin{aligned} f''(z) &= \lim_{h\to 0} \frac{f'(z+h) - f'(z)}{h} \\ &= \lim_{h\to 0} \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{h} \left[\frac{1}{(\zeta - z - h)^2} - \frac{1}{(\zeta - z)^2} \right] d\zeta \\ &= \lim_{h\to 0} \frac{1}{2\pi i} \int_C f(\zeta) \frac{2(\zeta - z) - h}{(\zeta - z - h)^2 (\zeta - z)^2} \, d\zeta. \end{aligned}$$

Again, the integrand converges uniformly on C to

$$\frac{2f(\zeta)}{(\zeta - z)^3}.$$

Interchanging the limit and the integral gives

$$f''(z) = \frac{2!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^3} \, d\zeta = \frac{2}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^3} \, d\zeta$$

similarly, we differentiate again to get

$$\begin{aligned} f'''(z) &= \lim_{h\to 0} \frac{f''(z+h) - f''(z)}{h} \\ &= \lim_{h\to 0} \frac{2!}{2\pi i} \int_C \frac{f(\zeta)}{h} \left[\frac{1}{(\zeta - z - h)^3} - \frac{1}{(\zeta - z)^3} \right] d\zeta. \end{aligned}$$

The difference quotient in the integrand converges uniformly on C to

$$\frac{3}{(\zeta - z)^4}.$$

hence

$$f'''(z) = \frac{3!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^4} \, d\zeta = \frac{6}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^4} \, d\zeta.$$

So by induction we get

$$\begin{aligned} f^{(n+1)}(z) &= \frac{n!}{2\pi i} \int_C \frac{(n+1) f(\zeta)}{(\zeta - z)^{n+2}} \, d\zeta \\ &= \frac{(n+1)!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+2}} \, d\zeta. \end{aligned}$$

Corollary. If $f : \Omega \to \mathbb{C}$ is a holomorphic function, and $\Omega \supseteq \overline{D_r}(z_0)$. Then f is smooth.

It turns out that life is even better than this! We know from MAT157 that we can talk about smooth functions which are infinitely differentiable, but we have an even better class. Namely analytic functions. So now we will define the notion of being analytic.

Definition. A (complex) **power series** is a formal sum of the form

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

Here z_0 denotes the center, everything here is complex!

Example. The **geometric series**

$$\sum_{n=0}^{\infty} z^n$$

The main thing to ask is when does this power series converge? There is the standard trick from calculus. Recall that if this series converges, then it must be equal to the limit

$$\lim_{N\to\infty} \sum_{n=0}^{N} z^n$$

We can use the telescoping method

$$\lim_{N\to\infty} \sum_{n=0}^{N} z^n = \lim_{N\to\infty} \frac{1 - z^{N+1}}{1 - z}$$

Let us ignore the case where $z = 1$ then we have

$$\frac{1 - \lim_{N\to\infty} z^{N+1}}{1 - z}$$

$$= \begin{cases} \frac{1}{1-z} & \text{if } |z| < 1 \\ \\ \text{Diverges if } |z| > 1 \end{cases}$$

For $z = 1$. The details are left as an exercise, The boundary case is treated separately.

Theorem. **(Radius of Convergence)** Given a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$. There exists an $R \in [0, \infty]$ such that

1. $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges absolutely if $|z - z_0| < R$
2. $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ diverges if $|z - z_0| > R$.

Moreover, $R = \frac{1}{L}$ where $L = \limsup_{n\to\infty} |a_n|^{1/n}$

Recall now that $\sum_{n=0}^{\infty} w_i$ where $w_i \in \mathbb{C}$ converges absolutely if $\sum_{n=0}^{\infty} |w_i| < \infty$.

Proof. Without loss of generality, assume that $z_0 = 0$ is the center. Now suppose that $|z| < R - \varepsilon$. Then we know that

$$\limsup_{n\to\infty} |a_n|^{1/n} = \frac{1}{R},$$

so

$$\limsup_{n\to\infty} |a_n|^{1/n}|z| = \frac{|z|}{R} < \frac{R - \varepsilon}{R} = 1 - \frac{\varepsilon}{R} \leq 1 - \delta$$

for some $\delta > 0$. This implies that there exists N such that for $n \geq N$, it follows that $|a_n|^{1/n}|z| \leq 1 - \delta$. Then test for absolute convergence

$$\sum_{n=0}^{\infty} |a_n||z|^n = \sum_{n=0}^{N} |a_n||z|^n + \sum_{n=N+1}^{\infty} |a_n||z|^n \leq C + \sum_{n=N+1}^{\infty} (1 - \delta)^n$$

Which proves the first case. Now, to prove the second case, we note that if $|z| > R$, then $|z| > R + \varepsilon$ for $\varepsilon > 0$

$$\limsup_{n\to\infty} |a_n|^{1/n} = \frac{1}{R}$$

so

$$\limsup_{n\to\infty} |a_n|^{1/n}|z| = \frac{|z|}{R} > \frac{R+\varepsilon}{R} = 1 + \frac{\varepsilon}{R} \geq 1 + \delta$$

for $\delta > 0$. This implies that there is a subsequence n_k such that $|a_{n_k}|^{1/n_k}|z| \geq 1 + \frac{\delta}{2}$ for any $k = 1, \ldots$. Implying that $|a_{n_k}||z|^{n_k} \geq \left(1 + \frac{\delta}{2}\right)^{n_k} \to \infty$. So the power series cannot converge. □

On Thursday, we will show that a holomorphic function can have a power series expansion. Moreover, we will also show another property of power series. It turns out that inside the disc, the convergence is really nice to the point we can differentiate and integrate term by term.

Theorem. If $f(z)$ is holomorphic in $D_R(z_0)$. Then we can write

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

Where R is the radius of convergence. $f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$. The power series also has radius of convergence R.

Power Series Strikes Back!

Last time, we proved Cauchy's integral formula, and as a consequence we know that holomorphic functions are smooth, even better, we will see that they are analytic! Recall that a (complex) power series is a sum of the form

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

where $a_n \in \mathbb{C}$ and z_0 is the center. We proved the radius of convergence theorem: Given a power series $\sum_{n=0}^{\infty} a_n (z-z_0)^n$. There exists an $R \in [0, \infty]$ such that

1. $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ converges absolutely if $|z - z_0| < R$
2. $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ diverges if $|z - z_0| > R$.

Moreover, $R = \dfrac{1}{L}$ where $L = \limsup_{n\to\infty} |a_n|^{1/n}$.

Similar to the case of real variables, we can differentiate a complex power series[1] term by term. If $f(z)$ is holomorphic in $D_R(0)$. Then we can write

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \tag{$\bigstar$}$$

Where R is the radius of convergence. Then the derivative also has a power series expansion, namely $f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$ in $D_R(0)$. This power series also has radius of convergence R.

Proof. First, we show that the power series of $f'(z)$ has the same radius of convergence as $f(z)$. To this end, $R = \dfrac{1}{\limsup_{n\to\infty} |a_n|^{1/n}}$ and $\limsup_{n\to\infty} n^{1/n} |a_n|^{1/n} = \limsup_{n\to\infty} |a_n|^{1/n}$. So the power series $\sum_{n=0}^{\infty} n a_n z^{n-1}$ has the same radius of convergence as ($\bigstar$). Let $g(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$. We claim that $g(z) = f'(z)$ for all $|z| < R$. Indeed, we write $f(z) = \sum_{n=0}^{N} a_n z^n + \sum_{n=N+1}^{\infty} a_n z^n$. More precisely,

$$f(z) = S_N(z) + E_N(z).$$

This implies that

$$f'(z) = \sum_{n=1}^{N} n a_n z^{n-1} + E_N'(z).$$

Claim: For $|z| < R$, then for any $\varepsilon > 0$, there exists N_1 such that whenever $N > N_1$, it follows that $|E_N'(z)| < \varepsilon$.

Proof of Claim. Note that

$$E_N'(z) = \lim_{h\to 0} \frac{E_N(z+h) - E_N(z)}{h}$$

[1] Here we assume it is centered at 0

So,

$$\begin{aligned}\left|\frac{E_N(z+h)-E_N(z)}{h}\right| &= \left|\sum_{n=N+1}^{\infty} a_n \frac{E_N(z+h)-E_N(z)}{h}\right| \\ &= \sum_{n=N+1}^{\infty} \left|\frac{E_N(z+h)-E_N(z)}{h}\right| |a_n| \\ &= \sum_{n=N+1}^{\infty} |a_n|\left(|z+h|^{n-1}+|z+h|^{n-2}|z|+\cdots+|z|^{n-1}\right) \\ &\le \sum_{n=N+1}^{\infty} |a_n| n |R-\varepsilon|^{n-1} \qquad \text{(if } h \ll 1 \text{ is sufficiently small)}\end{aligned}$$

Now since $R = (\limsup_{n\to\infty} |a_n|^{1/n})^{-1}$ this is a convergent sum, which means we can pick N_1 such that whenever $N > N_1$, it follows that $|E_N'(z)| < \varepsilon$. □

□

Now that our review of power series is over, we go back to holomorphic functions. Now, we want to show that if $f : \Omega \to \mathbb{C}$ then it is **analytic** if for any point $z_0 \in \Omega$, $f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$ in $D_r(z_0)$, $r > 0$. Analytic functions are so powerful, we can say that it gets as good as we want it to be in terms of smoothness, etc.

Theorem. Suppose that $f : \Omega \to \mathbb{C}$ is holomorphic, and $D_r(z_0) \subseteq \Omega$. Then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$$

for $|z - z_0| < r$

So as long as f is defined on the ball, then the radius of convergence can be bigger than r. The radius of convergence can get as big as it can get.

Example. Before we can prove this theorem, recall from last lecture that

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$$

for $|z| < 1$. So $f(z) = \frac{1}{(1-z)}$, then

$$\begin{aligned} f'(z) &= \frac{1}{(1-z)^2} \\ f''(z) &= \frac{2}{(1-z)^3} \\ &\vdots \\ f^{(n)}(z) &= \frac{n!}{(1-z)^{n+1}}\end{aligned}$$

More Consequences of CIF and Analytic Continuation

Last week we focused on Cauchy's integral formula, which states that for f a holomorphic function on $\Omega \to \mathbb{C}$. $\overline{D_r(z_0)} \subseteq \Omega$. Then for any $z \in D_r(z_0)$

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Where C is the boundary of $D_r(z_0)$ with the counterclockwise orientation. Moreover, we proved that we can differentiate this formula to obtain the following form for higher order derivatives:

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} \, d\zeta$$

Our goal now is to show that holomorphic functions are analytic, but first we start with a few basic applications:

Theorem. **(Cauchy's Inequality)** Suppose $f : \Omega \to \mathbb{C}$ is holomorphic, $\overline{D_r(z_0)} \subseteq \Omega$, and $C = \partial D_r(z_0)$. Then

$$\left|f^{(n)}(z_0)\right| \leq \frac{n!}{2\pi} \int_C \frac{|f(\zeta)|}{|\zeta - z_0|^{n+1}} |\mathrm{d}\ell| \leq \frac{n!}{2\pi} r^{-(n+1)} \, 2\pi r \sup_C |f| = \frac{n!}{r^n} \sup_C |f|.$$

Corollary. **(Liouville's Theorem)** Suppose that $f : \mathbb{C} \to \mathbb{C}$ is entire, namely f is holomorphic and well defined on all of $\mathbb{C}$, and is also bounded, then f is constant.

Proof of Liouville's theorem. Fix $z \in \mathbb{C}$ and suppose that $|f| \leq M$. Since f is entire, $\overline{D_R(z)} \subseteq \mathbb{C}$ for every $R > 0$. Cauchy's inequality for f' gives us

$$|f'(z)| \leq \frac{1}{R} \sup_{\partial D_R(z)} |f| \leq \frac{M}{R} \xrightarrow[R \to \infty]{} 0.$$

Hence $f' \equiv 0$, so f is constant. □

We discuss another important, and basic result which has to do with complex numbers. Recall how there is this dilemma in the real numbers where

$$x^2 + 1$$

has no real roots. So, the natural question is: Can the complex numbers guarantee roots for more general polynomials? The answer lies in the following theorem, which was first stated in MAT240

Theorem. **(Fundamental Theorem of Algebra)** Suppose $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ is a non-constant polynomial, where $a_i \in \mathbb{C}$. Then p has a root.

Proof. This is an application of Liouville's theorem. Suppose towards a contradiction that p is non-constant with no roots. Then $1/p$ is entire. We show it is bounded. Indeed, by the reverse triangle inequality,

$$|p(z)| \geq \underbrace{|a_n||z|^n - \cdots - |a_1||z| - |a_0|}_{\text{real polynomial in } |z|}.$$

Since $|a_n| \neq 0$ and $n \geq 1$, the right-hand side tends to ∞ as $|z| \to \infty$, so there is $R > 0$ with $|p(z)| \geq 1$ for $|z| > R$. On the compact set $\overline{D_R(0)}$ the continuous function $|p|$ is positive, hence attains a minimum $c > 0$. Therefore

$$|p(z)| \geq \min(1, c) > 0 \quad \text{for all } z \in \mathbb{C},$$

so $|1/p| \leq 1/\min(1, c)$ is bounded. By Liouville's theorem $1/p$ is constant, hence p is constant, contradicting that p is non-constant. □

This is a neat, complex analytic proof of this result, there are several other proofs of this. But we can see that lots of things follow by Cauchy's integral formula. Now, we want to talk about the analyticity of holomorphic functions. Recall that a function $f : \Omega \to \mathbb{C}$ is called (complex)-analytic if for any z_0 in Ω, we can write

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

in a neighborhood of z_0. We also stated the following theorem

Theorem. If $f : \Omega \to \mathbb{C}$ is holomorphic and $\overline{D_r(z_0)} \subseteq \Omega$, then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n \qquad \text{for } z \in D_r(z_0).$$

Proof. Let $C = \partial D_r(z_0)$ with the counterclockwise orientation. By the Cauchy integral formula,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} \, d\zeta$$

for any $z \in D_r(z_0)$. We want to write the integrand in terms of a power series. To this end, note that

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}}.$$

For $\zeta \in C$ we have $|\zeta - z_0| = r$, so $\left|\frac{z - z_0}{\zeta - z_0}\right| < 1$ is equivalent to saying $|z - z_0| < r$. Hence

$$\frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} = \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^n.$$

So we can write the Cauchy integral formula as

$$f(z) = \frac{1}{2\pi i} \int_C \left(\sum_{n=0}^{\infty} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} (z - z_0)^n \right) d\zeta.$$

We claim that we can exchange the sum and the integral to get

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, d\zeta \right) (z - z_0)^n.$$

To this end, it suffices to show that

$$\int_C \sum_{n=0}^{\infty} \frac{|f(\zeta)|}{|\zeta - z_0|^{n+1}} |z - z_0|^n \, \mathrm{d}\ell < \infty.$$

Let $M = \sup_C |f|$, which is finite since C is compact and f is continuous. Indeed, since $|\zeta - z_0| = r$ on C, note that

$$\frac{|f(\zeta)|}{|\zeta - z_0|^{n+1}} |z - z_0|^n \leq \frac{M}{r^{n+1}} |z - z_0|^n.$$

Since $|z - z_0| < r$, it follows that $\frac{|z - z_0|}{r} \leq 1 - \varepsilon$ for some $\varepsilon > 0$, and so

$$\sum_{n=0}^{\infty} \frac{|f(\zeta)|}{|\zeta - z_0|^{n+1}} |z - z_0|^n \leq \frac{M}{r} \sum_{n=0}^{\infty} (1 - \varepsilon)^n < \infty.$$

Since C has finite length, the integral above is finite, which justifies exchanging the sum and the integral. Finally, by the derivative form of the Cauchy integral,

$$\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta = \frac{f^{(n)}(z_0)}{n!},$$

so

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n,$$

concluding the proof. □

Remark: We used the Weierstrass M-test. (my goat)

This tells us that entire functions are also analytic, and they are convergent on any disc of convergence of any radius, so it has an infinite radius of convergence.

Example. Consider the exponential

$$e^z = \sum_{n=0}^{\infty} \frac{f^{(n)}}{n!} z^n = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

for any $z \in \mathbb{C}$. This agrees with the power series defined for the real exponential.

Now, we shift gears towards analytic continuation! This is another rigidity property of holomorphic functions.

Theorem. (Analytic Continuation) Suppose that $f : \Omega \to \mathbb{C}$ is holomorphic and Ω is connected. If there exists distinct $z_k \in \Omega$ such that $\lim_{k\to\infty} z_k = z_\infty \in \Omega$ and $f(z_k) = 0$ for all k. Then $f \equiv 0$ in Ω.

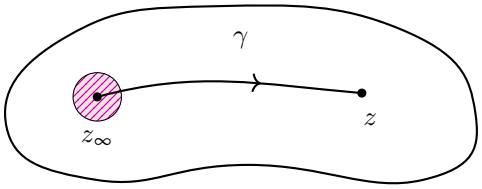
Proof. We have to harness the analyticity of f, suppose towards a contradiction that $f \not\equiv 0$. By analyticity, we get

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_\infty)^n$$

now, if all $a_n = 0$. Then f is 0 on a neighborhood of z_∞, and then connectedness gives us $f \equiv 0$, which is a contradiction. Hence, there exists a smallest k suich that $a_k \neq 0$. Moreover, we can write

$$f(z) = (z - z_\infty)^k \underbrace{\sum_{n=0}^{\infty} a_{n+k}(z - z_\infty)^k}_{g(z)}$$

Now, since $g(z_\infty)$, it follows from continuity that there is a neighborhood U of z_∞ on which $g(z) \neq 0$. Then $f(z) = 0$ for all $z \in U$. Implying that $z = z_\infty$. But $z_j \to z_\infty$, and each z_j is distinct. So, for large enough j, we have $z_j \in U$ where $z_j \neq z_\infty$, while $f(z_j) = 0$. Which again gives us a contradiction. Therefore f must be identically 0. □



We can use a different argument, since Ω is open and connected, we know it is path connected, so we show that f vanishes along γ in Ω. Indeed, let $t_0 = \inf\{t : f(\gamma(t)) \neq 0\}$, then take $t_k \nearrow t_0$ as $k \to \infty$. Now let $\widetilde{z_k} = \gamma(t_k) \to \widetilde{z_\infty} = \gamma(t_0)$. By the previous argument, we have that $f \equiv 0$ in $D_r(\widetilde{z_\infty})$ for some $r > 0$. Which is a contradiction.

Corollary. If $f, g : \Omega \to \mathbb{C}$ are holomorphic functions and Ω is open and connected. If $\lim_{k\to\infty} z_k = z_\infty \in \Omega$, $z_k \in \Omega$., and $f(z_k) = g(z_k)$. Then $f \equiv g$ in Ω.

Definition. Suppose that we have a holomorphic function $f : \Omega \to \mathbb{C}$, and $F : \Omega' \to \mathbb{C}$, is another holomorphic function and moreover, Ω, Ω' are open connected sets with $\Omega \subseteq \Omega'$. Then F is an **analytic continuation** of f if

$$F|_\Omega = f$$

Sequences of Holomorphic Functions

We saw some applications of Cauchy's integral formula, the most important of which was that holomorphic functions are analytic, unlike smooth real valued functions, which need not be analytic. The question that we want to discuss today is: what survives in taking limits of holomorphic functions? Say we take a sequence of holomorphic functions that converges to a limit; is the limit function holomorphic? In MAT157, we saw that if a sequence of differentiable functions converges uniformly, the limit function is continuous but need not be differentiable. We will see that in complex analysis, more properties are preserved.

Theorem. **(Morera's Theorem)** Suppose that $f : \mathbb{D} \to \mathbb{C}$ is continuous, and

$$\int_T f(z)\ dz = 0$$

for all triangles $T \subseteq \mathbb{D}$. Then f is holomorphic.

This is the converse of Goursat's theorem, so combining these two directions, we see that we can express holomorphic functions in terms of integrals. The proof follows roughly from everything we have done previously.

Proof. We summon the proof of Cauchy's theorem to obtain a primitive for f in D. That construction only uses $\int_T f\ dz = 0$ for triangles T, which is our hypothesis. So

$$F' = f \qquad (F : D \to \mathbb{C} \text{ is holomorphic}).$$

In particular, f is the derivative of a holomorphic function F. Since holomorphic functions have infinitely many complex derivatives, thanks to Cauchy's integral formula, $F' = f$ is itself holomorphic. □

A consequence of this is that the uniform limit of holomorphic functions is holomorphic. Namely, if $f_j : \Omega \to \mathbb{C}$ is a sequence of holomorphic functions and

$$\lim_{j\to\infty} \sup_{z\in\Omega} |f_j(z) - f(z)| = 0$$

for some $f : \Omega \to \mathbb{C}$, then f is holomorphic.

Note that f is continuous, since it is the uniform limit of continuous functions. Let $D \subseteq \Omega$ be a disc and $T \subseteq D$ a triangle.

By Goursat's theorem $\int_T f_j\ dz = 0$ for all j, and

$$\left|\int_T f\ dz\right| = \left|\int_T f\ dz - \int_T f_j\ dz\right| \leq \sup_T |f - f_j| \cdot \text{Length}(T) \xrightarrow[j\to\infty]{} 0.$$

That is,

$$0 = \int_T f_j\ dz \to \int_T f\ dz$$

so

$$\int_T f\ dz = 0.$$

By Morera's theorem f is holomorphic on D. The same argument works if $f_j \rightrightarrows f$ only on compact subsets, since each closed disc in Ω is compact.

Corollary. Power series are holomorphic inside their radius of convergence.

Proof. Note that

$$\sum_{n=0}^{\infty} a_n z^n = \lim_{N\to\infty} \sum_{n=0}^{N} a_n z^n$$

Point-wise in $D_R(0)$. We claim that this limit can be taken uniformly in $D_r(0)$[2] for any $r < R$. Indeed, one needs to show that the tail converges uniformly to 0 in $D_r(0)$. More precisely, we show that if $|z| < r$, then the tail $\sum_{n=N+1}^{\infty} a_n z^n \rightrightarrows 0.$ as $N \to \infty$

$$\begin{aligned}\left|\sum_{n=N+1}^{\infty} a_n z^n\right| &\le \sum_{n=N+1}^{\infty} |a_n||z|^n \\ &\le \sum_{n=N+1}^{\infty} |a_n| r^n\end{aligned}$$

Since $R = (\limsup_{n\to\infty} |a_n|^{1/n})^{-1}$ and $r < R$. We have that

$$\sum_{n=0}^{\infty} |a_n| r^n < \infty$$

implies

$$\sum_{n=N+1}^{\infty} |a_n| r^n \to 0$$

as $N \to \infty$ □

In fact, this tells us that being holomorphic is the same thing as analytic: holomorphic implies analytic by the previous lecture, and analytic implies holomorphic by the corollary above.

Now going back to the question posed earlier, we have that uniform limits of holomorphic functions are holomorphic. Moreover, unlike the real case, the derivatives of these also converge, as the next theorem shows.

Theorem. Suppose that $f_j : \Omega \to \mathbb{C}$ is a sequence of holomorphic functions, and $f_j \rightrightarrows f$ on compact subsets. Then

$$f_j' \rightrightarrows f'$$

on compact subsets.

Remark: This holds for all derivatives, namely $f_j^{(n)} \rightrightarrows f^{(n)}$ on compact subsets.

Proof. By the previous results f is holomorphic, so $f - f_j$ is holomorphic. By a covering argument (since every compact subset $K \subseteq \Omega$ is covered by finitely many discs $D_r(z_i)$ with $\overline{D_R(z_i)} \subseteq \Omega$ and $r < R$), it suffices to prove this on such discs. To this end, we fix z_0 with $\overline{D_R(z_0)} \subseteq \Omega$ and $r < R$. We need to bound the difference, let $\delta > 0$. Since $\overline{D_R(z_0)}$ is compact, there is J such that

$$|f_j - f| < \delta \quad \text{on } \overline{D_R(z_0)} \qquad \text{for all } j \ge J.$$

Let $r' = R - r$. For $z \in D_r(z_0)$ we have $\overline{D_{r'}(z)} \subseteq D_R(z_0)$. Cauchy's inequality for $f - f_j$ gives us for all $j \ge J$ and all $z \in D_r(z_0)$,

$$|f'(z) - f_j'(z)| \le \frac{1}{r'} \sup_{w\in\partial D_{r'}(z)} |f(w) - f_j(w)| \le \frac{\delta}{R - r}.$$

Since $\delta > 0$ was arbitrary, $f_j' \rightrightarrows f'$ on $D_r(z_0)$.

□

[2] We just assume that we are centered at 0

Zeros, Poles, and Meromorphic Functions

Some information:

1. Midterm on Tuesday (Exam Centre)
2. No homework, but there will be a practice midterm and problems
3. Zoom office hours on Monday
4. Similar in nature to homework (up to this week)

Last week, we discussed applications of Cauchy's integral formula (CIF), most importantly that holomorphic functions are analytic. Another application is the following: if f_k are holomorphic and $f_k \rightrightarrows f$ (uniformly on compact sets), then f is holomorphic. The same holds for the derivatives $f_k^{(j)}$.

We discuss singularities of holomorphic functions. The idea is to study the points where f is undefined. Near $z = 0$, consider

1. $\dfrac{e^z - 1}{z}$
2. $\dfrac{1}{z}$
3. $\dfrac{1}{z^2}$
4. $e^{1/z}$

Question: What can we say about the behavior of these functions near $z = 0$?

Answer:

1. For $\dfrac{e^z - 1}{z}$, it approaches a constant, so we can easily extend it using a Taylor series. Namely

$$\frac{e^z - 1}{z} = \frac{\left(1 + z + \frac{z^2}{2!} + \cdots\right) - 1}{z} = 1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots$$

   (We call these **removable singularities**.)

2. $\dfrac{1}{z}$ is not extendable holomorphically across $z = 0$. Indeed, if it were, then we could expand near $z = 0$ and the norm would be bounded, but it is not. So, we say that $\dfrac{1}{z}$ has a **pole** at $z = 0$, and it has order 1 (the number k such that $\left|\dfrac{1}{z^k}\right| \to \infty$).

3. $\dfrac{1}{z^2}$: same idea, but it has a pole of order 2 at $z = 0$.

4. $e^{1/z}$ is not a pole. Indeed, $x^N e^{1/x} \to \infty$ as $x \to 0^+$ (for every N). We call this an **essential singularity**.

Some definitions.

Definition. Suppose f is holomorphic on some domain. A point $a \in \mathbb{C}$ is an **isolated singularity** of f if f is defined on $D_r^*(a) = D_r(a) \setminus \{a\}$ for some $r > 0$.

We study these isolated singularities, and we classify them into 3 types

1. removable singularities
2. poles (tame singularities)
3. essential singularities (strange)

To study poles, we have to understand zeros of holomorphic functions (poles $\iff$ $\frac{1}{\text{zeros}}$).

Theorem. Suppose $f : \Omega \to \mathbb{C}$ is holomorphic and $f(a) = 0$, where f is not identically zero near a. Then there exists $m > 0$ such that we can write

$$f(z) = (z-a)^m h(z),$$

where $h(z)$ is holomorphic on Ω and $h(a) \neq 0$. Moreover, m is the **order** of the zero at a.

Remark: This is a local statement.

Proof. Look at the Taylor expansion of f at $z = a$. Since $f(a) = 0$ and f is not identically zero near a, there is a smallest $m > 0$ with $b_m \neq 0$, and

$$f(z) = \sum_{n=m}^{\infty} b_n (z-a)^n = (z-a)^m \underbrace{\sum_{n=0}^{\infty} b_{n+m}(z-a)^n}_{h(z)},$$

and then $h(a) = b_m \neq 0$. □

"Now we are ready to define what a pole is... not quite." - Freid Tong.

A consequence of this is that if f has a zero of order m at a, then locally in a neighborhood of a we have

$$c\,|z-a|^m \leq |f(z)| \leq C\,|z-a|^m$$

for some $c, C > 0$.

Definition. Let f be holomorphic and let a be an isolated singularity of f. Then

1. If f can be defined at $z = a$ such that the resulting function is holomorphic on $D_r(a)$ for some $r > 0$, then f has a **removable singularity** at a.
2. f has a **pole of order** m at a if we can write

$$f(z) = \frac{h(z)}{(z-a)^m} \quad \text{in some } D_r^*(a),$$

   where h can be extended to $D_r(a)$ and $h(a) \neq 0$.
3. f has an **essential singularity** at a if it is neither removable nor a pole.

Remark: This is equivalent to $\dfrac{1}{f}$ having a removable singularity at a, and having a zero of order m after extending.)

Theorem. **(Riemann's Theorem on Removable Singularities)** If $a \in \mathbb{C}$ is an isolated singularity of f, and if there exist $C > 0$, $r > 0$ such that $|f(z)| \leq C$ for $z \in D_r^*(a)$, then a is a removable singularity of f.

Proof. Consider the function

$$g(z) = \begin{cases} (z-a)^2 f(z) & z \neq a, \\ 0 & z = a. \end{cases}$$

Then, we claim that g is holomorphic on $D_r(a)$ for some $r > 0$.

Proof of claim: On $D_r^*(a)$, g is holomorphic as a product of holomorphic functions. At $z = a$, we have

$$g'(a) = \lim_{z \to a} \frac{g(z) - g(a)}{z - a} = \lim_{z \to a} (z-a) f(z) = 0,$$

as $|f| \leq C$ in $D_r^*(a)$.

This implies that g has a Taylor series expansion at $z = a$ in $D_r(a)$, where $a_0 = a_1 = 0$. So

$$\begin{aligned} g(z) &= a_2(z-a)^2 + a_3(z-a)^3 + \cdots \\ &= \sum_{n=2}^{\infty} a_n (z-a)^n \\ &= (z-a)^2 \sum_{n=0}^{\infty} a_{n+2}(z-a)^n. \end{aligned}$$

The series $\sum_{n=0}^{\infty} a_{n+2}(z-a)^n$ is equal to f on $D_r^*(a)$, and is well defined and holomorphic on $D_r(a)$. □

Definition. Given $\Omega \subseteq \mathbb{C}$ open, we say that f is **meromorphic** on Ω if there exists a countable collection $\{z_1, z_2, \dots\}$ (with no accumulation points in Ω) such that $f : \Omega \setminus \{z_1, z_2, \dots\} \to \mathbb{C}$ is holomorphic, and f only has poles at the z_i.

Exercise: Verify the following about meromorphic functions

1. Sums of meromorphic functions are meromorphic.
2. Products of meromorphic functions are meromorphic.
3. If $f \not\equiv 0$ is meromorphic, then $\dfrac{1}{f}$ is meromorphic.

Remark: Meromorphic functions (on a connected open set) form a field.

Example.

1. $\dfrac{1}{z}$
2. $\dfrac{1}{(z-a)^2}$

Theorem. **(Local Behavior of Meromorphic Functions)** If f has a pole of order m at a, then locally in

$D_r^*(a)$, f has a Laurent series expansion

$$f(z) = \sum_{n=-m}^{\infty} a_n (z-a)^n \qquad (a_{-m} \neq 0).$$

Remark: This is the same as

$$\frac{a_{-m}}{(z-a)^m} + \frac{a_{-m+1}}{(z-a)^{m-1}} + \cdots$$

Proof. Since f has a pole of order m at a, then

$$f(z) = \frac{h(z)}{(z-a)^m}, \qquad h(a) \neq 0,$$

and $h(z)$ is holomorphic on $D_r(a)$. This implies that

$$h(z) = \sum_{n=0}^{\infty} b_n (z-a)^n \quad \text{in } D_r(a),$$

and so

$$f(z) = \sum_{n=0}^{\infty} b_n (z-a)^{n-m}.$$

That is, $a_{n-m} = b_n$, and in particular $a_{-m} = b_0 = h(a) \neq 0$. □

Example. Let $f(z) = \dfrac{e^z}{z^3}$. This has a pole of order 3 about 0. Indeed, since $e^z = 1 + z + \dfrac{z^2}{2!} + \dfrac{z^3}{3!} + \cdots$, we have

$$f(z) = \underbrace{\frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{2!\,z}}_{\text{3 negative terms in Laurent series}} + \frac{1}{3!} + \cdots$$

Essential singularities are wild, there are many ways to quantitatively study this wildness.

Theorem. **(Casorati-Weierstrass)** If f has an essential singularity at a, then $f(D_r^*(a))$ is dense in $\mathbb{C}$ for all $r > 0$.

So the punctured disc $D_r^*(a)$ is mapped by f into $\mathbb{C}$, and we claim that its image meets every disc $D_\varepsilon(w)$.

Proof. We want to show for every $w \in \mathbb{C}$ and $\varepsilon > 0$, there exists $z \in D_r^*(a)$ such that $|f(z) - w| < \varepsilon$.

Assume towards a contradiction that there exist $w \in \mathbb{C}$, $\varepsilon > 0$ and $r > 0$ such that $f(D_r^*(a)) \cap D_\varepsilon(w) = \varnothing$. Let $h(z) = f(z) - w$. Then $|h(z)| \geq \varepsilon$ for all $z \in D_r^*(a)$, so h is nonvanishing there, $\dfrac{1}{h}$ is holomorphic on $D_r^*(a)$, and

$$\left|\frac{1}{h}\right| \leq \frac{1}{\varepsilon} \quad \text{in } D_r^*(a).$$

By Riemann's theorem, $\dfrac{1}{h}$ has a removable singularity at a. Then f has at most a pole at a. Contradiction. □

Residue Calculus

Last time, we classified (isolated) singularities of holomorphic functions:

(1) Removable (so fake).

(2) Poles (locally modeled around the singularity).

(3) Essential singularities (wild, neither (1) nor (2)).

We proved that if f has an essential singularity at a, then $f(D_r^*(a))$ is dense in $\mathbb{C}$ for all $r > 0$. This is the Casorati-Weierstrass theorem. Today, we discuss a Cauchy theorem for meromorphic functions. We ask the following question: How do poles contribute to the line integral of a meromorphic function around a closed contour C?

We start from the Cauchy integral formula. For $h : \Omega \supseteq \overline{D_r(z_0)} \to \mathbb{C}$ a holomorphic function, we have by the Cauchy integral formula

$$h(a) = \frac{1}{2\pi i} \int_C \frac{h(\zeta)}{\zeta - a}\, d\zeta$$

for $a \in D_r(z_0)$, where $C = \partial D_r(z_0)$.

Note that the integrand is a meromorphic function (of ζ) with at most a simple pole at $\zeta = a$.

Now, suppose that f is a meromorphic function on Ω with a pole at a. What is the integral of $f(z)\,dz$ along a closed (positively oriented) circle?

One way is to look at the Laurent series expansion around a. So

$$\begin{aligned} f(z) &= \sum_{n=-m}^{\infty} a_n (z-a)^n \\ &= \frac{a_{-m}}{(z-a)^m} + \frac{a_{-m+1}}{(z-a)^{m-1}} + \cdots + \frac{a_{-1}}{z-a} + a_0 + a_1(z-a) + \cdots \end{aligned}$$

This implies that for a small circle C (within the radius of convergence) we know that

$$\int_C f(z)\,dz = \sum_{n=-m}^{\infty} a_n \underbrace{\int_C (z-a)^n\,dz}_{= \begin{cases} 0 & \text{if } n \neq -1 \\ 2\pi i & \text{if } n = -1 \end{cases}} = 2\pi i\, a_{-1}.$$

Remark: The limit and integral swap is justified due to the uniform convergence in the radius of convergence.

This motivates the following definition.

Definition. We define the **residue** of f at a to be

$$\operatorname{Res}(f, a) = a_{-1}.$$

Example.

(1) $\operatorname{Res}\left(\frac{1}{z}, 0\right) = 1$.

(2) $\operatorname{Res}\left(\frac{1}{z^2}, 0\right) = 0$.

(3) $\operatorname{Res}\left(\frac{e^z}{z^2}, 0\right)$. Note that

$$\frac{e^z}{z^2} = \frac{1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots}{z^2} = \frac{1}{z^2} + \frac{1}{z} + \cdots$$

So $\operatorname{Res}\left(\frac{e^z}{z^2}, 0\right) = 1$.

Going back to what we set up, the residue theorem generalizes CIF to arbitrary meromorphic functions with many more poles and over any contour.

Theorem. **(Residue Theorem)** If $f : \Omega \to \mathbb{C}$ is meromorphic on Ω, and $\Gamma \subseteq \Omega$ is a (positively oriented) toy contour enclosing a region $D \subseteq \Omega$, and f has no poles on Γ, then

$$\int_\Gamma f(z)\, dz = 2\pi i \sum_{k=1}^{n} \operatorname{Res}(f, a_k),$$

where $a_1, \ldots, a_n$ are the poles of f in D.

Proof. The idea is to subtract off the singular parts of f near each pole. Near each pole a_k, we have the Laurent expansion

$$\begin{aligned} f(z) &= \sum_{n=-m}^{\infty} b_{n,k}(z - a_k)^n \qquad \text{(holomorphic in a punctured neighborhood of } a_k) \\ &= \sum_{n=-m}^{-1} b_{n,k}(z - a_k)^n + \sum_{n=0}^{\infty} b_{n,k}(z - a_k)^n \\ &= \underbrace{P_k(z)}_{\text{principal part}} + \sum_{n=0}^{\infty} b_{n,k}(z - a_k)^n. \end{aligned}$$

Consider now

$$h(z) = f(z) - \sum_{k=1}^{n} P_k(z),$$

which is holomorphic on D (the singularities at each a_k are removable). Implying that

$$\int_\Gamma h(z)\, dz = 0,$$

and so

$$\begin{aligned}\int_\Gamma f(z)\,dz &= \sum_{k=1}^{n} \int_\Gamma P_k(z)\,dz \\ &= \sum_{k=1}^{n} \int_\Gamma \frac{b_{-1,k}}{z-a_k}\,dz \\ &= \sum_{k=1}^{n} \operatorname{Res}(f,a_k) \int_\Gamma \frac{dz}{z-a_k} \\ &= \sum_{k=1}^{n} \operatorname{Res}(f,a_k)\, 2\pi i.\end{aligned}$$

Here, the third line uses $b_{-1,k} = \operatorname{Res}(f,a_k)$, and the second line uses that $\int_\Gamma (z-a_k)^n\,dz = 0$ for $n \neq -1$. □